

Self-Play Can Mask Frontier LLM Risk Responses in AI Development Races

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Abstract

Self-play is a common way to evaluate language-model agents, but it entangles a focal endpoint with an adapting copy of itself. A measured response to risk may therefore describe the interaction rather than the endpoint alone. We test this problem in an idealised AI-development race, where faster UNSAFE development raises the winner’s chance of losing its payoff. Nine commercial endpoints faced a frozen task-validity screen; five named higher-tier endpoints entered the strategic analysis.

In self-play, all five reduce UNSAFE play as stated risk rises, although four respond gradually and one switches from the ceiling to the floor. We then hold the rival fixed by code. With ten matched repetitions per endpoint, rival and risk condition, the paired Always-Unsafe minus Always-Safe rival contrast is positive in all 15 endpoint-by-risk cells, and every 95% interval excludes zero. More importantly, the paired rival-by-risk interaction is endpoint-specific: the rival contrast grows with risk for Claude Opus 5 and GPT-5.5, shrinks for Claude Sonnet 5, and is unresolved for Gemini 3 Flash and GPT-5.4. Self-play also exceeds the fixed-safe rate in 13 of 15 cells. Thus a mirror-match rate cannot be read as a context-independent risk preference.

Two supporting checks sharpen that conclusion. Four graded endpoint profiles are closer in shape to a weak-selection evolutionary benchmark than to its strong-selection setting. Repeated samples from the same eligible endpoint also occupy fewer first-five-round trajectories than every matched draw of different human participants in 13 of 15 cells. Evaluation should therefore fix and audit the opponent, and report trajectories as well as aggregate rates. Claims remain scoped to the tested endpoints, one prompt version and one idealised game.

CCS Concepts

• Computing methodologies → Multi-agent systems.

Keywords

Large language models, AI development races, multi-agent systems, repeated games, AI safety, opponent-conditioned behaviour

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1 Introduction

Self-play is attractive because it is symmetric and easy to reproduce. It is also a compound measurement: the evaluated endpoint is both the focal agent and the environment to which that agent reacts. When a treatment changes behaviour, the observed rate may therefore reflect two adapting copies rather than a stable property of either one. This matters for language models used in strategic settings, where responses to an opponent can be exactly what the evaluation is trying to distinguish from responses to risk [2, 16, 20].

We study this identification problem in an idealised race over AI development. Each round, two agents choose between slower SAFE development and faster UNSAFE development; UNSAFE creates a delayed chance that the winner loses its payoff. The game represents a controlled speed–safety dilemma rather than a real company in full [4, 7]. A broader AI-race literature models bidding incentives, heterogeneous actors, regulation and voluntary safety commitments [8, 13–15]; scenario games likewise identify coordination as central to race outcomes [12]. Its value is experimental: stated danger, rival policy and progress can be held fixed separately. We reproduce the two-player mechanism of Fernández Domingos and Han [11]. One hosted model at one fixed configuration is an *endpoint*; model families can differ across dated versions and provider settings.

LLM behavioural research now spans cognitive benchmarks and repeated matrix games [2, 9], reproducible testbeds with language and payoff interventions [6, 18, 19], and multi-agent common-resource societies [24]. Outcomes can change sharply when the interaction partners change [20], and fixed opponents are established in LLM game experiments [2, 22]. Our question is narrower: *does replacing the mirror opponent with a fixed, undisclosed policy alter or mask the magnitude and shape of the response to the stated catastrophe risk itself?* That is the estimand a safety reading needs. A high or low self-play rate alone cannot answer it.

Two checks protect this inference. First, a valid action format does not prove that an endpoint tracked state, arithmetic or terminal scoring. We use a frozen operational inclusion rule and let the game engine, rather than generated text, resolve every transition. Second, even equivalent prompt renderings can change model output in strategic games [21, 25]. More generally, formatting and answer order matter [23, 27, 28], as do small prompt and persona changes [16, 26, 29]. We therefore separate confirmatory gameplay from presentation diagnostics and retain the prompt, raw response, parsed action, state and terminal record for every decision.

The answer is not one universal opponent effect. In self-play, every included endpoint reduces UNSAFE play as stated risk rises, with four graded profiles and one threshold-like switch. Against four fixed scripted rivals, however, all five endpoints move substantially with the rival. A paired interaction analysis then shows three distinct shapes: the rival effect grows with risk on Claude Opus 5 and GPT-5.5, shrinks on Claude Sonnet 5, and is unresolved on

Gemini 3 Flash and GPT-5.4. Self-play can therefore alter or conceal the treatment-response curve, not merely shift its level.

This is the paper’s core contribution: a matched, replay-audited demonstration that a self-play risk response in this game is an opponent-conditioned outcome. The human and evolutionary analyses are supporting checks, not separate novelty claims. Human-proxy studies report both aggregate fidelity and distributional or behavioural distortions [1, 3, 5]. We therefore test whether aggregate agreement hides distributional mismatch, using the source study’s first-five-round trajectories [11]; the latter asks whether the observed curve depends on the evolutionary selection regime used as a reference. An exploratory group-size study remains in the supplementary material because it covers one endpoint and cannot identify a pure group-size effect.

Every behavioural conclusion is scoped to its tested endpoint, prompt version, decoding contract, provider path and run period. The labels SAFE and UNSAFE denote actions inside the experimental game and are not measurements of general LLM safety, of a subjective risk preference, or of suitability for autonomous deployment, and nothing measured here is a harm, a real safety outcome, or an action taken by an organisation.

2 Preliminaries

We first specify the two-player game reproduced from Fernández Domingos and Han [11] (§2.1). We additionally evaluate an exploratory N -player extension; its mechanism and results are reported in the supplementary material. Figure 1 summarises the two-player instrument and the three empirical lenses used in the main results.

2.1 Two-player game

Two companies $i \in \{1, 2\}$ play a repeated race. In every round t , both simultaneously choose an action $a_i^t \in \{S, U\}$ (Safe, Unsafe) and commit to it before either learns the other’s choice, and only then does the round resolve (Figure 1a). Resolution has two parts. First, progress accumulates as $P_i^t = P_i^{t-1} + \sigma(a_i^t)$ with $\sigma(S) = 1.0$ and $\sigma(U) = 1.5$, so Unsafe always closes the race faster. Second, the round pays a stage payoff $\pi_i^t = \pi(a_i^t, a_{-i}^t)$, read off the fixed matrix (own action by row, opponent action by column, Figure 1c):

$$\pi = \begin{pmatrix} 1.0 & 0.6 \\ 2.4 & 2.0 \end{pmatrix}. \quad (1)$$

Unsafe strictly dominates Safe at the stage level (row 2 exceeds row 1 in both columns): the only cost of Unsafe development enters later, through accumulated private risk.

The race lasts a minimum of 5 rounds; after every completed round from round 5 onward it terminates with probability 0.2, giving an expected horizon $\mathbb{E}[T] = 9$. Neither company is ever told the horizon, so neither can condition on the current round being the last (Figure 1a). Termination resolves in two further steps. First, the company with the higher cumulative progress P_i^T wins a prize $B = 100$, split 50/50 on an exact tie. Second, and only for that winner or tied winners, an accumulated private setback risk is drawn:

$$q_i(T) = p_r^{\max} \cdot \frac{n_i^U(T)}{T}, q_i(T) \in [0, 1], p_r^{\max} \in \{0.10, 0.60, 0.90\}, \quad (2)$$

Table 1: The four benchmark strategies. Each is fixed by its opening move and by what it does afterwards, so AS and AU never react to the other player while CS and CAS react to nothing else. These four are also the fixed rivals of §4.2.

Strategy	Opening	Every later round
Always Safe (AS)	SAFE	SAFE
Conditionally Safe (CS)	SAFE	copy the rival’s last move
Conditionally Antisocial Safe (CAS)	UNSAFE	copy the rival’s last move
Always Unsafe (AU)	UNSAFE	UNSAFE

where $n_i^U(T)$ counts company i ’s Unsafe choices over the whole race and $p_r^{\max}$ is a treatment-level constant, assigned once per race, that we vary across three risk levels. If the draw against $q_i(T)$ fails, that company’s entire race payoff, accumulated stage payoffs plus the prize, is zeroed. A company that ends behind is never subject to the draw: it keeps its stage payoffs regardless of how much Unsafe it played, but receives no share of the prize (Figure 1a). This asymmetry, in which risk is built by both companies but realised only against whoever is ahead, is what makes Unsafe development a genuine gamble rather than a free speed advantage.

2.2 Game-theoretic benchmark

The two incentives oppose each other. Stage-level dominance rewards Unsafe immediately, while across the full race Unsafe choices raise the setback risk for a winner or tied winner, so the best action can depend on the risk condition, earlier actions, and the expected response of other players.

We use four simple strategies as a theory benchmark, defined in Table 1. Two of them ignore the other player and two of them do nothing but answer it, which is what makes the set a benchmark: it separates a fixed disposition from a reaction. In the reconstructed finite-population evolutionary model of Fernández Domingos and Han [11], the most common strategy changes from AU at $p_r^{\max} = 0.10$, to CAS at 0.60, and to CS at 0.90: the theory predicts not one fixed amount of Unsafe play but a risk-dependent shift between unconditional speed, an aggressive opening and conditional restraint. We use this as a qualitative benchmark, not as a fitted model of LLM output, because prompted self-play trajectories are not samples from an evolutionary population. Selection strength β controls how sharply payoff differences change strategy frequencies: strong selection can create an abrupt switch, while weak selection produces a more graded response. Mutation keeps alternative strategies present. These are shape references, not mechanisms attributed to the endpoints.

3 Experimental design and evidence policy

3.1 Agent protocol and units of analysis

At each decision an agent receives a versioned prompt carrying the round, the assigned maximum private risk, the disclosed state, and the preceding revealed action profile when $t > 1$. Same-round decisions are sealed: every response comes from the same pre-action snapshot, before the engine resolves actions, progress, payoffs and risk, so the environment rather than generated text determines all transitions. Raw response, parsed action, retry and parser status,

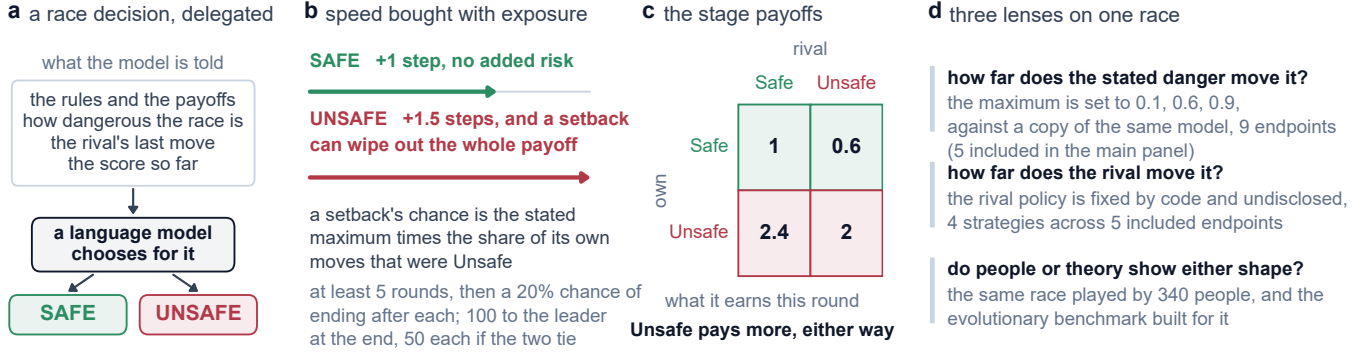


Figure 1: The game and the questions it lets us ask. (a) The model receives the rules, stated danger, rival’s last move and current score, then returns either SAFE or UNSAFE. (b) SAFE advances one step without added risk; UNSAFE advances faster but raises the chance that the winner’s payoff is wiped out. (c) The two-player stage-payoff matrix makes UNSAFE attractive against either rival move, so the delayed terminal risk supplies the safety tension. (d) The main results examine stated-risk response, scripted-rival response, and comparison with human trajectories and the evolutionary benchmark.

pre-turn state, terminal record, configuration and seed allocation are retained for every decision.

Sampling parameters here are provider-visible contracts, not intended settings. The run manifests record a requested temperature of 0.7 that the hosted SDK forwarded on no baseline endpoint, and no provider confirmed an effective value. Seed handling was endpoint-specific: the SDK stripped the requested seed on the Google endpoints and forwarded it, application unconfirmed, on the OpenAI and Anthropic ones.

Both seats of a race are the same endpoint, so every gameplay rate below is mirror-match play, until §4.2 fixes the rival instead. For endpoint m , stated risk p and rival policy π , define

$$R_m(p, \pi) = \Pr(\text{UNSAFE} \mid p, \pi). \quad (3)$$

Self-play observes only $R_m(p, m)$. The scripted-rival campaign instead compares $R_m(p, \pi)$ across policies π fixed by code, and its rival-by-risk estimand is the change from low to high risk in the paired Always-Unsafe minus Always-Safe contrast.

The main outcome is whether an agent chooses UNSAFE. A race is the independent unit because later decisions depend on earlier states. We pool decisions only when reporting a cell’s rate. For comparisons, we first calculate the difference within each matched repetition and then average those differences, so every repetition receives equal weight. A *parse failure* is a response outside the required action format; because the fallback changes later states, it marks the whole race as contaminated.

3.2 Evidence strata and admission

We separate confirmatory gameplay from diagnostic presentation checks. A *diagnostic pilot* is an exploratory test for possible effects or failures; it does not estimate a stable effect. A *confirmatory* study tests a question under a plan fixed before any result is inspected, with its protocol, endpoint, prompt, configuration, race count and exclusion rules recorded in the run manifest. We never pool the two strata.

One presentation check is confirmatory and is described in the supplementary material: on two endpoints, a fully crossed rerun

retells the same game under a race story or as an abstract contest, and substitutes the opaque response codes P and Q for SAFE and UNSAFE under both code mappings. A task-validity screen gates which endpoints enter the behavioural comparison. It is an inclusion rule, not a behavioural outcome, and all gameplay results below are analysed without ranking endpoints by its scores.

3.3 Quality-control screen on the endpoints

An action that parses is not evidence that the agent followed the race. Before behavioural analysis, each endpoint therefore answered the same frozen bank of 20 probes over six task-validity domains, three times. An endpoint entered the study only if it cleared the fixed thresholds of 80% overall accuracy, 75% state reconstruction and 75% terminal scoring. Five of nine endpoints passed; the other four still played the same confirmatory baseline and remain available throughout the trajectory-diversity analyses for transparency and sensitivity checks.

The bank and operational thresholds were fixed before the campaign, and the screen reads probes rather than races, so gameplay cannot change its verdict. It was not preregistered: three of nine routes received their first verdict after their gameplay had been collected. Two admitted endpoints clear one screening domain at the minimum attainable passing score, and the admitted/refused split exactly matches the absence/presence of mini, nano or lite in the route name. The cutoffs are not externally validated. We therefore use the screen only as an operational inclusion rule, never as a measure of comprehension, capability or behaviour. The supplement retains every route and score. Within the observed stability box, the admitted set is unchanged over all 676 combinations of the three achievable thresholds. The boundary is nevertheless visible: lowering state reconstruction by one answer, from the 75% rule to the next attainable score of 73.3%, admits Gemini 3.1 Flash Lite. This sensitivity is why the screen supports inclusion rather than a capability ranking.

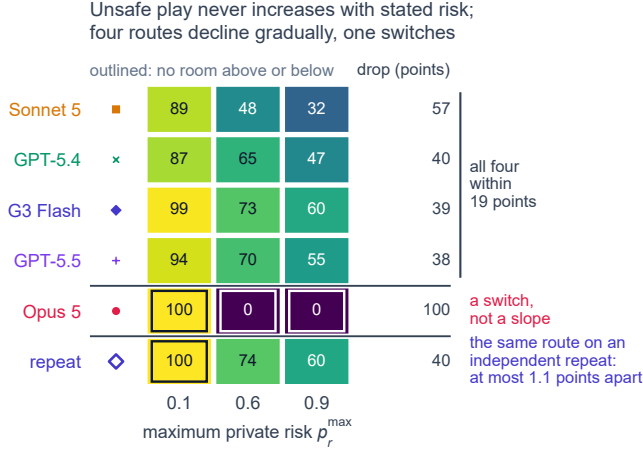


Figure 2: UNSAFE play is non-increasing with stated risk. The four graded endpoints fall within a 19-point band; Claude Opus 5 switches from the ceiling to the floor. The outlined cells are at a boundary and therefore leave no room for a larger or smaller observed rate. The repeat row is the same Gemini 3 Flash endpoint collected under an independent administration.

4 Results

Self-play first supplies the baseline that the scripted-rival campaign reinterprets. The main result then tests whether the apparent risk response survives when the opponent is fixed. Evolutionary and human comparisons serve as supporting checks on response shape and trajectory concentration. A screen-failing endpoint appears below only where it is explicitly named as excluded; full tables and robustness checks are in the supplement.

4.1 Self-play suggests a shared directional risk response

There is a shared directional response across the eligible endpoints. UNSAFE play is non-increasing with the stated maximum private risk on every eligible endpoint, while the shape varies: four of the five give ground gradually and the fifth switches (Figure 2). Each endpoint supplied 30 races and 558 dependent within-race decisions under one frozen protocol, with zero parse failures and ten races per risk cell. Risk contrasts use matched repetition blocks, and race-weighted reanalysis preserves the low-to-high-risk contrast on every eligible endpoint (supplementary material).

The four graded endpoints, spanning three providers, have risk-response drops within a 19-point band, from 38.2 to 57.0 percentage points. We report that band rather than an average over all five, because averaging a switch with four slopes returns 54.8 points, a figure no graded endpoint is near and which one row produces.

That fifth endpoint follows a threshold-like policy. Claude Opus 5 plays UNSAFE on all 186 decisions at risk 0.1 and SAFE on all 372 decisions above it, with no race inside a cell departing from that. Its 100-point figure reaches the largest value on the scale, and because no risk level was run between 0.1 and 0.6, the location of its transition is not measured.

Table 2: Presentation contrasts in percentage points, paired within 60 repetition blocks. Positive values mean that the second presentation raises UNSAFE play. Brackets are 95% bootstrap intervals.

Endpoint	Contrast	Effect [95% interval]
Gemini 3 Flash	code <i>Q</i> means SAFE	+9.5 [+5.6, +13.8]
Claude Sonnet 5	code <i>Q</i> means SAFE	+8.3 [+5.4, +12.2]
Gemini 3 Flash	abstract contest	+10.8 [+6.0, +15.6]
Claude Sonnet 5	abstract contest	+3.4 [+0.9, +7.4]

At the highest risk the game offers, where a winner’s whole payoff is zeroed with probability up to 0.9 in proportion to how much UNSAFE it played, the four graded endpoints still play UNSAFE on roughly a third to three fifths of their decisions against a copy of themselves, in the $r=0.9$ column of Figure 2. This variation shows that risk-aware behaviour remains contingent on endpoint and interaction context. Two endpoints can also reach a similar rate through very different sets of races, which §4.4 measures directly.

For the repeated Gemini 3 Flash endpoint, the profile was stable across two administrations. Repeating the whole protocol on that endpoint under the same frozen contract moved no risk cell by more than 1.1 points over 186 decisions, against a 93.0-point spread in risk response across the nine endpoints, and moved its own risk response from 38.7 to 40.3. This bounds run-to-run variation for that endpoint at this sample size; it licenses no such bound for an endpoint collected once, which matters here because no provider confirmed applying the requested sampling seed.

Representation-equivalent prompt changes also move rates. The directional risk response survives a fully crossed narrative-by-symbol rerun on two endpoints, but neither the action-code mapping nor the narrative skin is behaviourally inert (Table 2). The similar mapping effects replicate across the two tested providers; the different story effects warn against treating two endpoints as a population of models.

4.2 Fixed rivals reveal opponent-conditioned risk responses

In the five endpoints tested against scripted rivals, the observed behaviour does not support a context-independent risk profile. Across these endpoints, changing who is on the other side moves play substantially, but the magnitude varies by endpoint and can overlap the effect of changing how dangerous the race is told to be. Every rate reported so far is mirror-match play, in which the rival is the endpoint. One campaign fixes the rival instead.

The rival in that campaign is code, not a model. Five eligible endpoints each played the four benchmark strategies of §2.2 at all three risk levels: 60 cells, ten races each, 600 races and 5,580 endpoint decisions with no parse failure anywhere. It is never disclosed to the endpoint, and every one of its 5,580 recorded rival moves was recomputed afterwards from the endpoint’s own recorded history, which no cell failed. The endpoint’s seat is counterbalanced five and five inside each cell, and each contrast is taken inside a repetition, so the hidden horizon is differenced out. Because rival-policy assignment is fixed and exogenous within matched blocks, the direction of a difference between rival policies is identified. Conditional policies



Figure 3: Changing the scripted rival changes UNSAFE play on every eligible endpoint. (a) Tiles show rates against four rivals at each risk level. (b) Larger tiles show the paired Always-Unsafe minus Always-Safe effect; every corresponding 95% interval excludes zero. The right column gives the paired change in that effect from risk 0.1 to 0.9; solid outlines exclude zero and dashed outlines do not. Comparisons are within endpoints because endpoint settings differ.

Table 3: Does stated risk change the rival effect? Entries are the paired interaction $[(AU - AS)_{0.9} - (AU - AS)_{0.1}]$ in percentage points, with 95% intervals over ten matched repetitions. Positive values mean that changing the rival matters more at high risk.

Endpoint	Rival $\times$ risk interaction
Gemini 3 Flash	-1.8 [-7.9, 3.7]
GPT-5.4	+2.3 [-4.6, 9.4]
GPT-5.5	+36.4 [29.0, 44.0]
Claude Sonnet 5	-18.9 [-25.1, -11.8]
Claude Opus 5	+29.2 [20.9, 36.4]

can still respond mechanically to the endpoint’s previous move, as specified by the fixed policy.

All five tested endpoints respond to the scripted opponent policy. UNSAFE play rises from the rival that always plays SAFE, through the two conditional rivals, to the rival that never plays SAFE, in fourteen of fifteen endpoint-by-risk cells, with strict ordering in twelve. The one reversal is GPT-5.5 at risk 0.1, where the CAS rival draws 100.0% and Always Unsafe is 97.8%, so this is a boundary case rather than a clean counterexample (Figure 3a). The paired difference between the two unconditional rivals is positive in every cell. All fifteen paired-block 95% intervals have positive lower bounds, with the smallest at +16.5 points (Figure 3b). The response to the stated risk survives the change of design as well: against a fixed safe rival, no endpoint’s UNSAFE rate rises as the stated risk rises.

The interaction in Table 3 applies Equation 3 directly to test whether the apparent risk response depends on the rival. The rival contrast grows from low to high risk on Claude Opus 5 and GPT-5.5, shrinks on Claude Sonnet 5, and is statistically unresolved on Gemini 3 Flash and GPT-5.4. There is therefore no universal “risk amplifies rivalry” pattern; the shape itself is endpoint-specific. This conclusion is within endpoint. Decoding contracts differ, so effect-size comparisons across endpoints remain descriptive. With only ten matched repetitions per endpoint-by-risk cell, the intervals are necessarily coarse: they resolve three directional interactions and leave two unresolved, rather than demonstrating that the latter two are zero.

The two conditional rivals make the dependence visible at the first move. They follow the same policy from round two onward and differ only in whether the rival opens SAFE or UNSAFE. That opening-move contrast has a positive lower 95% bound in twelve of fifteen endpoint-by-risk cells, showing that one exogenous rival action can redirect the later path.

A mirror-match rate is therefore an interaction outcome of one policy meeting itself, rather than a stand-alone measurement of how an endpoint treats risk. For Gemini 3 Flash, the self-play profile is 98.9%, 73.1% and 60.2%, whereas the same endpoint against Always Safe gives 24.7%, 17.2% and 14.0%. Across the full campaign, self-play exceeds the fixed-safe rate in thirteen of fifteen cells. The two exceptions are Claude Opus 5 at the floor: 0.0% against 1.1% at risk 0.6, and a 0.0% tie at risk 0.9.

The bound on the finding is the design itself. Four reduced strategies are four points in a space of policies rather than a sample of

rivals, so this establishes what an endpoint does against *these* rivals, on one game and one prompt version, and five commercial endpoints are not a sample from a population of models.

4.3 A weak-selection benchmark matches the graded shape

The four graded endpoint profiles are closer to the source study’s human-fitted weak-selection regime than to its strong-selection reference setting (Figure 4b). The direct profiles remain the clearest shape comparison: Gemini 3 Flash gives 98.9%, 73.1% and 60.2%, and Claude Sonnet 5 gives 89.2%, 48.4% and 32.3% over 30 races each. At the reference setting of Fernández Domingos and Han [11], $\beta = 2$ and $\mu = \beta/Z$ lose 98 points inside a risk step of 0.005 (Figure 4a). The all-five fit shows that Claude Opus 5 is the qualitative exception: its profile is near-step rather than graded, even though its weak-selection RMSE is still lower than its reference RMSE.

In the evolutionary benchmark, the abrupt transition arises under strong selection and becomes graded under weak selection. We swept selection strength across $\beta \in [0.001, 10]$ crossed with three mutation rules, 90 stationary-distribution cells, each estimated from independent chains with the between-chain spread retained so that a badly mixed cell cannot be mistaken for a result. As β falls towards neutrality the predicted high-risk UNSAFE rate rises off the floor and the model’s own risk response becomes graded, which is the shape the endpoints show. At $\beta = 0.01$ with $\mu = 0.05$ the model predicts 87.3%, 63.9% and 38.0%, the closest cell in the entire sweep to both direct endpoints, within 9.6 percentage points of Claude Sonnet 5 and 15.4 of Gemini 3 Flash in root-mean-square error.

The weak-selection cell was not selected solely to fit these endpoints: $\beta = 0.01$ and $\mu = 0.05$ are also the values the source study reports as its own best fit to human play [11]. The interpretation stays bounded. A stationary distribution describes strategy frequencies in a population under selection and mutation, while our measurements are prompted decisions in self-play; the agreement in shape says the graded risk response is consistent with weak selection and high exploration, not that the endpoints are evolving. What the sweep establishes is narrower and still useful: a claim that LLM agents contradict the evolutionary benchmark cannot rest on one setting of the selection strength, because the sign of the high-risk gap depends on it.

4.4 Repeated endpoint samples occupy fewer paths than human samples

Samples from different human participants span many paths; repeated samples from one fixed endpoint often reuse far fewer. An endpoint can therefore match a human average without reproducing the human sample’s heterogeneity. We compare exact first-five-round joint-action trajectories, the longest prefix observed in every race under the minimum-five-round horizon, represented as ten-bit strings. The confirmatory baseline contributes 60 trajectories per endpoint. The only human reference is the de-identified dataset of Fernández Domingos and Han [11], so every comparison is bounded to that population and game.

Of 341 released participants, 340 have complete five-round paths. The primary comparison uses the 336 participants whose partner is also complete, forming 168 dyads. A separate reconstruction check

of the source study’s dynamic model uses 338 participants and 2,888 round-level observations (supplement).

Figure 5 compares each endpoint sample with 20,000 equally sized resamples of ten complete human dyads at each risk. Sampling whole dyads preserves the interaction, and equal sample sizes prevent the larger human dataset from gaining an automatic diversity advantage. Eighteen of 27 endpoint cells, including 13 of the 15 eligible cells, fall below every dyad draw. The participant-level sensitivity gives 21 of 27, including all 15 eligible cells; the primary dyad-null floors are 12, 15 and 13 distinct trajectories at risks 0.1, 0.6 and 0.9. The complementary participant-level table records effective counts of 18.9, 19.7 and 19.3 and mean pairwise distances of 0.47, 0.50 and 0.50. Panel b sorts the observed paths by frequency and panel c displays the modal five-round joint-action sequence. The same cells support two numerical checks reported in the supplement: Hill $q=1$ counts effective trajectories [17], and mean pairwise Hamming distance measures separation of their ten action bits. Under that complementary Hill $q=1$ analysis, every eligible endpoint sits entirely below the human interval at every risk level.

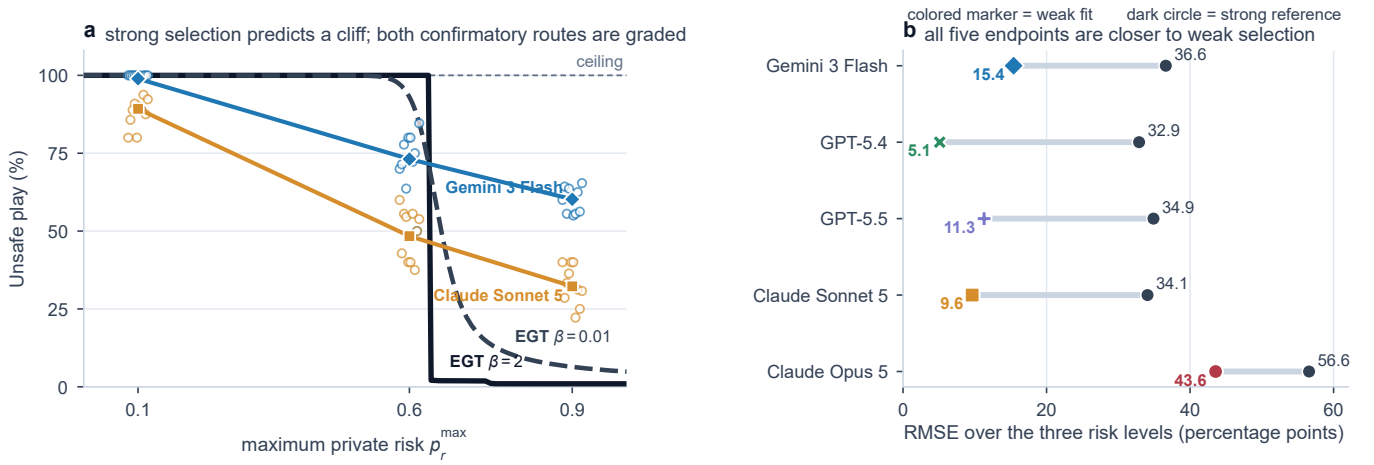
The concentration is strongest for Claude Opus 5, which produces a single trajectory, repeated by all 20 of its trajectories, at every risk level, while Claude Sonnet 5 uses two at risk 0.1. Concentration is also distinct from level: Gemini 3 Flash and GPT-5.5 finish within five points of each other at the highest risk and use a similar number of distinct trajectories, 12 and 13, yet the Gemini trajectories sit half again as far apart from one another, 0.41 against 0.27 in mean pairwise distance.

Under the participant-level sensitivity, two endpoints are the only exceptions, and the screen refuses both of them. GPT-5.4 mini uses 20, 17 and 17 distinct trajectories out of 20 at risks 0.1, 0.6 and 0.9, and GPT-5.4 nano 16, 19 and 16; both land inside the participant-level human null at every risk level, and no other endpoint reaches it once under that sensitivity. On this measure they are not less diverse than people. Under the primary dyad null, GPT-5.4 and GPT-5.5 are not below the finite-draw floor at risk 0.9, so the cluster-aware headline is 13 of 15 eligible cells rather than all 15. The two participant-sensitivity exceptions are outside the eligible set fixed by the inclusion rule of §3.3 before any race was read, so they are reported, not interpreted as strategic diversity.

5 Discussion and conclusion

These agents incorporate both stated risk and the rival’s stance. UNSAFE play is non-increasing with risk on every eligible endpoint, but fixing the rival reveals three different interactions: the rival effect grows with risk on two endpoints, shrinks on one and is unresolved on two. The self-play curve is therefore not an endpoint-only risk profile.

Two measurement rules follow. To estimate how behaviour depends on the opponent, fix the rival by code and audit every move. To describe heterogeneity, report paths as well as aggregate rates. In 13 of 15 eligible cells, repeated samples from one endpoint occupy fewer distinct histories than every matched draw of different human participants. Claude Opus 5 is the limiting case, with one trajectory in each condition.



Panel a shows the two prespecified direct profiles; panel b uses the canonical secondary fit for all five eligible endpoints.

Figure 4: The evolutionary benchmark and the eligible endpoint profiles in a common risk coordinate. (a) The direct comparison keeps the two prespecified profiles, Gemini 3 Flash and Claude Sonnet 5, with individual race rates behind their means. The small-mutation limit is step-like at strong selection and graded near weak selection, while both direct profiles move gradually. (b) Each dumbbell compares the canonical RMSE over the three matched risk levels for the reference strong-selection regime ($\beta = 2$, $\mu = .02$) and the reported best weak-selection regime ($\beta = .01$, $\mu = .05$). All five eligible endpoints are closer to the weak-selection fit; Claude Opus 5 remains the qualitative near-step exception in panel (b).

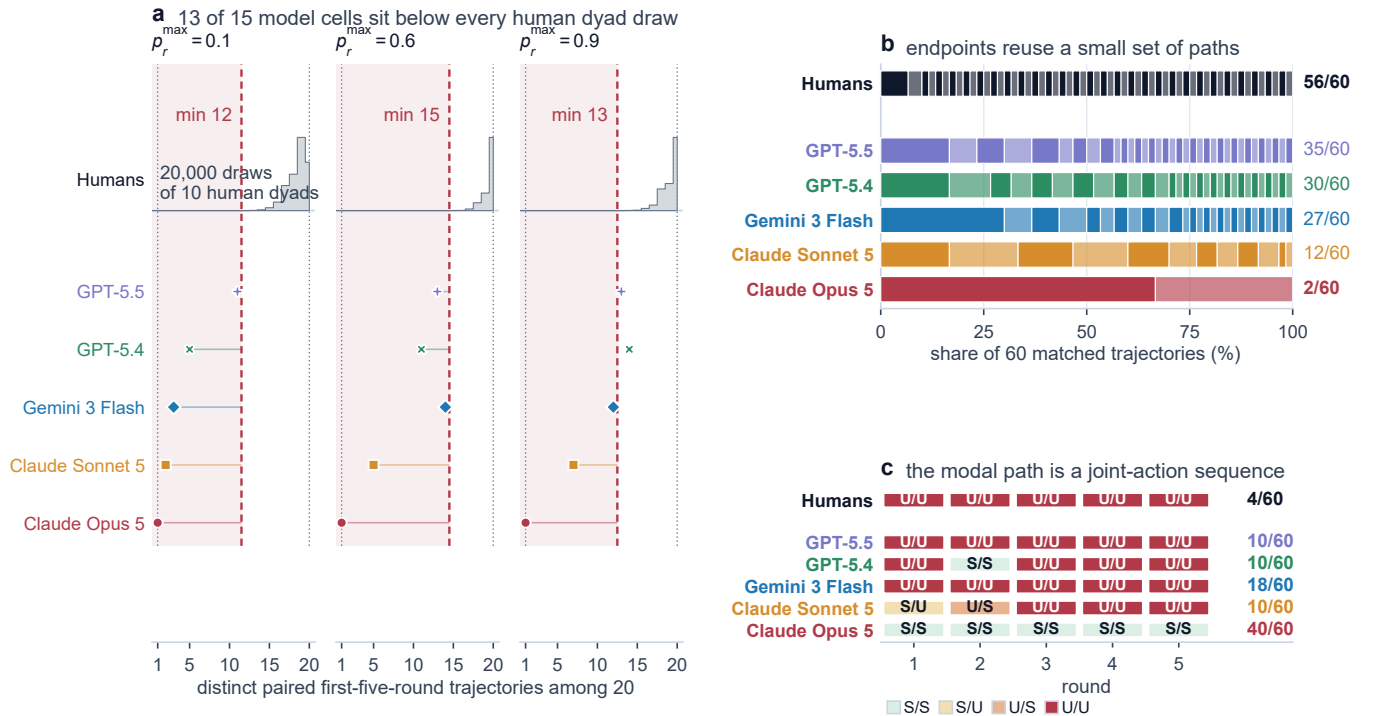


Figure 5: Repeated samples from eligible endpoints usually occupy fewer paths than matched human dyad draws. (a) Distinct first-five-round paths; the dashed line is the minimum in 20,000 matched draws (13/15 cells fall below it). (b) Frequency-ranked paths over 60 trajectories. (c) The modal joint-action sequence. A participant-level sensitivity places all 15 cells below every draw.

The human experiment is an external behavioural reference, not a target an agent should reproduce. Where an endpoint matches a human association, the two populations behave similarly on that one measured aspect of this task; it does not follow that they share a mechanism, and a mismatch does not invalidate the human result.

The scope is deliberately narrow. Nine commercial endpoints are not a model population, cross-endpoint contracts differ, and the scripted campaign covers five endpoints against four chosen policies. The human reference is one dataset from this game, while the evolutionary reference is a reconstruction checked against pinned EGTtools source [10], not a fit of the endpoints. Presentation controls cover two endpoints, the group-size study covers one, and provider-side temperature and seed handling were not confirmed. Finally, the game omits collective harm, regulation, communication and uncertainty about real capabilities. These limits rule out claims about general model safety, intent or real organisations.

The next discriminating tests are clear: repeat the matched rival-by-risk design at larger sample size, include adaptive rivals with declared policies, and apply the same trajectory measure in a second game. Those tests would show which interactions persist beyond this prompt, endpoint roster and mechanism.

Code, the de-identified run records and a checker that recomputes every reported quantity from them are provided with the supplementary material and will be released publicly with the final version.

The main lesson is simple: self-play rates are interaction outcomes. Evaluations of strategic agents should vary the opponent and compare whole trajectories, not only aggregate choices.

References

- [1] Gati V Aher, Rosa I. Arriaga, and Adam Tauman Kalai. 2023. Using Large Language Models to Simulate Multiple Humans and Replicate Human Subject Studies. In *Proceedings of the 40th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 202)*, Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (Eds.). PMLR, Honolulu, Hawaii, USA, 337–371. <https://proceedings.mlr.press/v202/aher23a.html>
- [2] Elif Akata, Lion Schulz, Julian Coda-Forno, Seong Joon Oh, Matthias Bethge, and Eric Schulz. 2025. Playing repeated games with large language models. *Nature Human Behaviour* 9, 7 (01 Jul 2025), 1380–1390. doi:10.1038/s41562-025-02172-y
- [3] Lisa P. Argyle, Ethan C. Busby, Nancy Fulda, Joshua R. Gubler, Christopher Rytting, and David Wingate. 2023. Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis* 31, 3 (2023), 337–351. doi:10.1017/pan.2023.2
- [4] Stuart Armstrong, Nick Bostrom, and Carl Shulman. 2016. Racing to the precipice: a model of artificial intelligence development. *AI & SOCIETY* 31, 2 (01 May 2016), 201–206. doi:10.1007/s00146-015-0590-y
- [5] James Bisbee, Joshua D. Clinton, Cassy Dorff, Brenton Kenkel, and Jennifer M. Larson. 2024. Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis* 32, 4 (2024), 401–416. doi:10.1017/pan.2024.5
- [6] Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. 2025. FAIRGAME: A Framework for AI Agents Bias Recognition Using Game Theory. In *ECAI 2025 (Frontiers in Artificial Intelligence and Applications, Vol. 413)*. IOS Press, 4097–4104. doi:10.3233/FAIA251300
- [7] Stephen Cave and Seán S. ÓhÉigeartaigh. 2018. An AI Race for Strategic Advantage: Rhetoric and Risks. In *Proceedings of the 2018 AAAI/ACM Conference on AI, Ethics, and Society* (New Orleans, LA, USA) (AIES '18). Association for Computing Machinery, New York, NY, USA, 36–40. doi:10.1145/3278721.3278780
- [8] Theodor Cimpeanu, Francisco C. Santos, Luís Moniz Pereira, Tom Lenaerts, and The Anh Han. 2022. Artificial intelligence development races in heterogeneous settings. *Scientific Reports* 12, 1 (02 Feb 2022), 1723. doi:10.1038/s41598-022-05729-3
- [9] Julian Coda-Forno, Marcel Binz, Jane X. Wang, and Eric Schulz. 2024. CogBench: A Large Language Model Walks into a Psychology Lab. In *Proceedings of the 41st International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 235)*. PMLR, Vienna, Austria, 9076–9108. <https://proceedings.mlr.press/v235/coda-forno24a.html>
- [10] Elias Fernández Domingos. 2025. EGTtools: Toolbox for Evolutionary Game Theory. Zenodo. doi:10.5281/zenodo.16237051 Software.
- [11] Elias Fernández Domingos and The Anh Han. 2026. Falling Behind Drives Unsafe Development in an Idealised AI Race Experiment. arXiv:2607.26034 [cs.AI] <https://arxiv.org/abs/2607.26034>
- [12] Ross Gruetzmacher, Shahar Avin, James Fox, and Alexander K Saeri. 2025. Strategic insights from simulation gaming of AI race dynamics. *Futures* 167 (2025), 103563. doi:10.1016/j.futures.2025.103563
- [13] The Anh Han, Tom Lenaerts, Francisco C. Santos, and Luis Moniz Pereira. 2022. Voluntary safety commitments provide an escape from over-regulation in AI development. *Technology in Society* 68 (2022), 101843. doi:10.1016/j.techsoc.2021.101843
- [14] The Anh Han, Luís Moniz Pereira, and Tom Lenaerts. 2019. Modelling and influencing the AI bidding war: a research agenda. In *Proceedings of the 2019 AAAI/ACM Conference on AI, Ethics, and Society*. Association for Computing Machinery, New York, NY, USA, 5–11. doi:10.1145/3306618.3314265
- [15] The Anh Han, Luís Moniz Pereira, Francisco C. Santos, and Tom Lenaerts. 2020. To Regulate or Not: A Social Dynamics Analysis of an Idealised AI Race. *Journal of Artificial Intelligence Research* 69 (2020), 881–921. doi:10.1613/jair.1.12225
- [16] Nathan Herr, Fernando Acero, Roberta Raileanu, María Pérez-Ortiz, and Zhibin Li. 2024. Are Large Language Models Strategic Decision Makers? A Study of Performance and Bias in Two-Player Non-Zero-Sum Games. arXiv:2407.04467 [cs.AI] <https://arxiv.org/abs/2407.04467>
- [17] Mark O. Hill. 1973. Diversity and Evenness: A Unifying Notation and Its Consequences. *Ecology* 54, 2 (1973), 427–432. doi:10.2307/1934352
- [18] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2025. Understanding LLM Agent Behaviours via Game Theory: Strategy Recognition, Biases and Multi-Agent Dynamics. arXiv:2512.07462 [cs.MA] doi:10.48550/arXiv.2512.07462 Preprint.
- [19] Trung-Kiet Huynh, Duy-Minh Dao-Sy, Thanh-Bang Cao, Phong-Hao Le, Hong-Dan Nguyen, Phu-Quy Nguyen-Lam, Minh-Luan Nguyen-Vo, Hong-Phat Pham, Phu-Hoa Pham, Thien-Kim Than, Chi-Nguyen Tran, Huy Tran, Gia-Thoai Tran-Le, Alessio Buscemi, Le Hong Trang, and The Anh Han. 2026. Payoff Scaling Shapes Cooperation in LLM Agents across Languages. arXiv:2601.19082 [cs.AI] doi:10.48550/arXiv.2601.19082 Preprint, version 2 revised June 2026.
- [20] Paraskevas V. Lekeas and Giorgos Stamatopoulos. 2026. How a Cooperative-Override Circuit Suppresses Nash Play in Large Language Models. arXiv:2604.27167 [cs.GT] <https://arxiv.org/abs/2604.27167>
- [21] Seyed Pouyan Mousavi Davoudi, Arshia Gharagozlou, Alireza Amiri-Margavi, Amin Gholami Davodi, and Hamidreza Hasani Balyani. 2026. Same Game, Different Story: A Minimal Conservative Strategic Robustness Benchmark for Large Language Model Agents. arXiv:2607.19670 [cs.MA] <https://arxiv.org/abs/2607.19670> Preprint.
- [22] Kenneth Payne and Baptiste Alloui-Cros. 2025. Strategic Intelligence in Large Language Models: Evidence from evolutionary Game Theory. arXiv:2507.02618 [cs.AI] <https://arxiv.org/abs/2507.02618>
- [23] Pouya Pezeshkpour and Estevam Hruschka. 2024. Large Language Models Sensitivity to the Order of Options in Multiple-Choice Questions. In *Findings of the Association for Computational Linguistics: NAACL 2024*. Association for Computational Linguistics, Mexico City, Mexico, 2006–2017. doi:10.18653/v1/2024.findings-naacl.130
- [24] Giorgio Piatti, Zhijiang Jin, Max Kleiman-Weiner, Bernhard Schölkopf, Mrinmaya Sachan, and Rada Mihalcea. 2024. Cooperate or Collapse: Emergence of Sustainable Cooperation in a Society of LLM Agents. In *Advances in Neural Information Processing Systems*, Vol. 37. Curran Associates, Inc., Red Hook, NY, USA. doi:10.52202/079017-3548
- [25] Isaac Robinson and John Burden. 2025. Framing the Game: How Context Shapes LLM Decision-Making. arXiv:2503.04840 [cs.CL] <https://arxiv.org/abs/2503.04840>
- [26] Abel Salinas and Fred Morstatter. 2024. The Butterfly Effect of Altering Prompts: How Small Changes and Jailbreaks Affect Large Language Model Performance. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 4629–4651. doi:10.18653/v1/2024.findings-acl.275
- [27] Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. 2024. Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I Learned to Start Worrying About Prompt Formatting. In *International Conference on Learning Representations*, Vol. 2024. OpenReview.net, Vienna, Austria, 25055–25083. https://proceedings.iclr.cc/paper_files/paper/2024/hash/6c0e99d736da621403018ca7b32b1a4d-Abstract-Conference.html
- [28] Sheng-Lun Wei, Cheng-Kuang Wu, Hen-Hsen Huang, and Hsin-Hsi Chen. 2024. Unveiling Selection Biases: Exploring Order and Token Sensitivity in Large Language Models. In *Findings of the Association for Computational Linguistics: ACL 2024*. Association for Computational Linguistics, Bangkok, Thailand, 5598–5621. doi:10.18653/v1/2024.findings-acl.333
- [29] Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. 2024. When “A Helpful Assistant” Is Not Really Helpful: Personas in System Prompts Do Not Improve Performances of Large Language Models. In *Findings of the Association for Computational Linguistics: EMNLP 2024*. Association for Computational Linguistics, Miami, Florida, USA, 15126–15154. doi:10.18653/v1/2024.findings-emnlp.888